

Dr. RICHARD H. TAYLOR

CURRICULUM VITAE ET STUDIORUM

(2026)

Richard H. Taylor, Ph.D.

Scientist-Engineer • Educator • Entrepreneur • Investor • Artist and Poet

Ph.D., Mechanical Engineering & Materials Science — Duke University

Salt Lake City, Utah, USA

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Google Scholar: scholar.google.com/citations?user=N6dbY7QAAAAJ

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Summary: Ph.D. in Mechanical Engineering & Materials Science (Duke, 2013) with twenty+ publications and conference proceedings spanning high-throughput *ab initio* materials discovery, computational thermodynamics, failure analysis, and materials education. Foundational contributor to the AFLOW framework and the AFLOWLIB.org consortium—co-author of the field’s two highly-cited works (AFLOW and AFLOWLIB.ORG, 1,800+ and 1,300+ citations respectively) and lead author of the AFLOWLIB.org RESTful API. Visiting Lecturer and Visiting Researcher at MIT (2014–2017); instructor and course developer at MIT; Guest Scientist at NIST (2012–2014) on the Materials Genome Initiative. As entrepreneur and operator: CEO of Oaki and Woven Pear, co-founder of Fresh Farms (AFS exclusive brand), and co-founder of TP Investments in multi-family and industrial real estate. Materials and product consultant to nine and ten-figure organizations including Teck Resources, Werner Ladder, Jacuzzi, Trex, Purple Mattress, Enso Rings, and Tracker Connect. Vice-Chair of the Peterson School of Music; developed strategy for its capital campaign. Pushcart-nominated poet and visual artist with work exhibited in galleries and museums across Utah and collected nationally.

1. CURRENT WORK

- **Periodic Graph Metamaterials Atlas (Aux-Mat)** 2025 – present
Project Originator, aux-mat.org.
 - Project on the mechanical and dynamical properties of textile metamaterials: periodic mesh structures whose topologies are derived from textile-construction patterns.
 - Released v0.1.0 of the atlas characterizing 321 lattice grounds from the TesseLace bobbin-lace catalog, screened through static elasticity (spring and beam models), phonon dispersion, and humidity-driven swelling.
 - Developed a two-dimensional classification of mechanical response distinguishing load-driven auxeticity from swelling-driven contraction.
 - Atlas data and code released open-source (CC-BY 4.0 / MIT). Preprint in preparation.
- **The Consonance Tree** ongoing
Creator, r-h-taylor.github.io/consonance-tree.
 - Interactive web tool for visualizing and sounding harmonic trajectories on the Stern–Brocot tree of rational numbers; plays and transforms just-intonation sequences and chords.
 - Companion to the paper *A Structure of Harmonic Relations* (in preparation).
- **Eagar Corpus — Materials Engineering: A Narrative Pedagogy** ongoing
Editor, r-h-taylor.github.io.

- Scholarly digital edition of the recorded MIT teaching of Prof. Thomas W. Eagar (1949–2022): 215 lectures and 1,949 referenced cases.
- Four-layer, full-text-searchable corpus with editorial apparatus and methodology; methods paper in preparation.

2. EDUCATION

- **Duke University**, Durham, NC, USA.
Ph.D., Mechanical Engineering & Materials Science, 2013.
Dissertation: *Crystallographic Symmetry Algorithms in a High-Throughput Framework*.
Advisor: Prof. Stefano Curtarolo.
- **Duke University**, Durham, NC, USA.
M.S., Mechanical Engineering & Materials Science, 2012.
Thesis: *A High-Throughput Framework for Materials Research and Space Group Determination Algorithm*.
- **Brigham Young University**, Provo, UT, USA.
M.S., Physics, 2010.
Thesis: *Prediction and Study of Binary Alloys Using First-Principles Methods*.
- **Brigham Young University**, Provo, UT, USA.
B.S., Physics, 2007. Thesis: *Logistic Maps and Synchronization*.

3. ACADEMIC AND RESEARCH POSITIONS

- **Massachusetts Institute of Technology (MIT)**, Cambridge, MA.
Visiting Lecturer and Research Affiliate, Department of Materials Science & Engineering. 2014–2017
 - Taught *Structure and Properties of Materials* (3.072) to undergraduate engineering students.
 - Co-taught and developed content for *Simulation and Modeling*. with Professor Eugene Fitzgerald
 - Developed curriculum, course materials, and examinations for introductory materials science coursework.
 - Contributed to MITx/edX online course development, expanding the reach of MIT’s engineering curriculum to global learners.
 - Postdoctoral research on failure mechanisms in flexible water hoses, corrugated stainless steel tubing, and fire ignition systems; collaborated on hydrogen generation via surface activation of bulk aluminum. Materials characterization using SEM, metallography, and optical microscopy. PI: Prof. Thomas W. Eagar.
- **National Institute of Standards and Technology (NIST)**, Gaithersburg, MD. 2012–2014
Guest Scientist.
 - Team Lead, first-principles data infrastructure design and development for the U.S. Materials Genome Initiative.
 - Team Lead, DFT benchmarking and error-analysis project.
 - Developed crystallographic symmetry algorithms and contributed to computational materials frameworks.
- **Duke University**, Durham, NC. 2013–2014
Postdoctoral Associate, Department of Mechanical Engineering & Materials Science.
 - Developed the high-throughput materials data RESTful API for the AFLOWLIB.org consortium—used internationally by the materials-research community.
 - Contributed to the AFLOW source code for automated materials discovery (www.aflowlib.org).

- Research focus: crystallographic symmetry, structure–property relationships, binary alloy systems (Mg, Mg–Li, Pt–Ti and related families).
- **Duke University**, Durham, NC. 2010–2013
Instructor and Course Developer / Graduate Researcher, Department of Mechanical Engineering & Materials Science.
 - Developed and taught curriculum for an introductory materials-science course; created exams and lab materials.
 - Performed research on HT materials discovery
 - Contributed to the AFLOW codebase using cpp
- **Brigham Young University**, Provo, UT. 2007–2010
Teaching and Research Assistant, Department of Physics & Astronomy.
 - Delivered physics lab and lecture instruction for undergraduate non-major and major courses.
 - Tutored first- and second-year physics and mathematics students.
 - Led group instruction for calculus-based physics (mechanics, thermodynamics, electromagnetism).
 - Supervised physics labs for both non-major and numerically oriented lab courses.
 - Provided individual and small-group instruction across first- and second-year physics and mathematics.
 - Research on logistic maps, synchronization phenomena, and coupled-map dynamics (collaborator: Prof. Gus L. W. Hart).

4. INDUSTRY AND ENTREPRENEURIAL LEADERSHIP

- **Outdoor Adventure Kids** 2019 – present
CEO/operator, www.oaki.com.
 - Lead product development, materials strategy, and brand positioning for a consumer-goods company in children’s outerwear and lifestyle products.
 - Direct go-to-market, wholesale, and direct-to-consumer channels.
 - Apply materials-selection frameworks and engineering principles to drive product innovation across textiles, polymers, and printed substrates.
 - Apply materials and product design principles to reduce carbon impact of product line, eliminated use of PFAS DWR treatments, implemented full life-cycle recycling program.
- **Woven Pear** ongoing
Co-CEO / Owner, www.wovenpear.com.
 - End-to-end product development and go-to-market strategy for a designed-graphic sock and apparel brand.
 - Oversee product design, sourcing, manufacturing, brand marketing, and wholesale/retail distribution.
- **Fresh Farms** (exclusive brand developed for Associated Food Stores) 2017–2019
Co-Founder.
 - Directed brand and product development from concept to retail launch as an exclusive private-label program for AFS.
 - Built product portfolio, packaging system, and retail execution strategy.
 - Executed a global sourcing strategy, working directly with growers and producers in South Africa and beyond.
- **Sunné Kombucha**, 2017–2018
Advisor / Operator: brand development and wholesale-sales channel strategy with successful multi-store regional launch.

5. REAL ESTATE INVESTMENT AND DEVELOPMENT

- **TP Investments** present
 - Co-founder. Manages active real-estate property development.
 - Specializes in multi-family residential and industrial real-estate development.
 - Responsibilities include acquisition due diligence, capital structuring, contractor selection, materials and finish specification, and long-term portfolio strategy.

6. PRODUCT AND MATERIALS CONSULTING

Independent and retained consulting engagements in product development, materials selection, and failure analysis for engineering and consumer-goods clients across multiple sectors. 2014–2020

Selected clients:

- **Teck Resources Limited** — materials and product development consulting.
- **Werner Ladder** — engineering and materials consulting.
- **Jacuzzi** — product development and materials consulting.
- **Trex Company** — composite materials and product development.
- **Purple Mattress** — materials and product development.
- **Enso Rings** — elastomer and consumer-product development.
- **Tracker Connect** — product development consulting.

Engagement scope: failure analysis, root-cause investigations, materials-selection frameworks, design-for-manufacture review, and competitive product benchmarking.

Litigation Engineering Expert — technical analysis of failed systems, components, and devices.

- Catastrophic failure analysis on matters involving \$100M+ losses.
- Expert witness testimony and technical report development.
- Successful deposition record.
- Root-cause investigations spanning mechanical, materials, and design-defect issues.

7. TEACHING EXPERIENCE

- **MIT**, Department of Materials Science & Engineering 2014–2017
 - *Structure and Properties of Materials* (3.072) — lecturer, exam and content developer.
 - *Simulation and Modeling* — contributor.
 - *MITx/edX* — content and curriculum developer for global online learners.
- **Duke University**, Department of Mechanical Engineering & Materials Science 2011–2014
 - Developed and taught curriculum for an introductory materials-science course; authored exam and lab materials.
- **Brigham Young University**, Department of Physics & Astronomy 2007–2010
 - Lab and lecture instruction for undergraduate physics (non-majors and majors).
 - Tutoring across first- and second-year physics and mathematics.
 - Group instructor for calculus-based physics (mechanics, thermodynamics, electromagnetism).
 - Supervised physics labs for non-major and numerically oriented lab sections.
 - One-on-one and small-group tutoring across first- and second-year physics and mathematics curricula.

8. RESEARCH INTERESTS

- High-throughput *ab initio* materials discovery and computational thermodynamics.
- Crystallographic symmetry algorithms; space-group determination; structure prototyping.
- Binary and ternary alloy phase stability (Mg-, Li-, Pt-, Ti-, Hf-based systems).
- Materials informatics, RESTful APIs for scientific data exchange, and open data infrastructure (AFLOWLIB.org).
- Failure analysis of engineered systems (flexible hoses, corrugated tubing, fire-ignition systems).
- Materials selection and design pedagogy; engineering education.
- Materials engineering and science historiography.
- Applied materials and product development for consumer goods.

9. TECHNICAL SKILLS AND COMPETENCIES

Programming and computation: C++, Python, R, MATLAB, Mathematica, Maple, Bash/Linux, L^AT_EX.

Computational materials: density functional theory (DFT), high-throughput *ab initio* calculations, crystallographic algorithms, AFLOW framework, RESTful API design.

Materials characterization: SEM, metallography, optical microscopy.

Fabrication & prototyping: 3D printing, machining, prototyping workflows.

Engineering domains: materials science, failure analysis, mechanical engineering, product design, thermodynamics, structural materials.

Curriculum development: course design, exam writing, online/edX content authoring, lab instruction.

Business and operations: product development, brand development, go-to-market strategy, wholesale and DTC sales, financial planning, governance and board operations.

Governance: board governance and committee leadership, endowment design and capital campaigns, executive search and succession planning, grant-review process consulting, and policy and bylaw development for nonprofit and founder-led organizations.

Real Estate and Construction: Active contractor's license in the State of Utah. Experience spans multi-family development, permitting, re-zoning, municipal incentive-based development programs, construction practice, and design.

10. PEER-REVIEWED PUBLICATIONS, PREPRINTS, AND CONFERENCE PROCEEDINGS

[Articles listed in approximate reverse-chronological order. Citation counts as listed on Google Scholar, May 2026.]

[2026, in preparation / under review]

23. R.H. Taylor and N. Jenkins, *Thermal Decomposition Morphology of Alkyd Paint on Steel: A Heat-Direction Diagnostic for Fire Scene Reconstruction*, manuscript in preparation (2026).
22. R.H. Taylor, *Emergent Fibonacci Structure from Local Growth Rules*, manuscript in preparation (April 2026).
21. R.H. Taylor, *Periodic Graph Metamaterials Atlas: a topology screen for periodic 2D stick lattices derived from bobbin lace*, arXiv preprint, in preparation (2026). aux-mat.org.

[2026]

20. R. Taylor, *The Governance Gap: We Trust Executives and Micromanage Everyone Else*, SSRN (2026). DOI: [10.2139/ssrn.6608238](https://doi.org/10.2139/ssrn.6608238).

[2020]

19. J.T. Slocum, T.W. Eagar, R. Taylor, and D.P. Hart, *Activation of bulk aluminum and its application in a hydrogen generator*, *Applied Energy* **279**, 115712 (2020). [29 citations.]

18. C. Toher, C. Oses, D. Hicks, E. Gossett, F. Rose, P. Nath, D. Usanmaz, et al. (incl. R.H. Taylor), *The AFLOW fleet for materials discovery*, in Handbook of Materials Modeling: Methods, Theory and Modeling, Springer (2020), pp. 1785–1812. [40 citations.]
- [2018]**
17. D. Hicks, C. Oses, E. Gossett, G. Gomez, R.H. Taylor, C. Toher, M.J. Mehl, et al., *AFLOW-SYM: platform for the complete, automatic and self-consistent symmetry analysis of crystals*, Acta Crystallographica Section A: Foundations and Advances **74**(3), 184–203 (2018). [73 citations.]
- [2017]**
16. R.H. Taylor, H.R. Larson, and T.W. Eagar, *Electric Arc Holes in Corrugated Stainless Steel Tubing*, Fire Technology **53**(5), 1919–1932 (2017). DOI: [10.1007/s10694-017-0663-1](https://doi.org/10.1007/s10694-017-0663-1).
- [2016]**
15. R.H. Taylor and T.W. Eagar, *Flexible Water Hose Failures: A Case Study and General Design Considerations*, Journal of Failure Analysis and Prevention **16**(3), 333–336 (2016). [2 citations.]
14. J. Slocum, D. Hart, T. Eagar, and R. Taylor, *Practical Method of Hydrogen Generation by Surface Activation of Bulk Aluminum*, (2016).
13. N. Jenkins, R. Taylor, and T. Eagar, *Nanoscale Size Effect on Thermodynamic Stability of Hexavalent Chromium in Welding Fume*, FABTECH 2016, American Welding Society, Las Vegas, NV.
- [2014]**
12. R.H. Taylor, F. Rose, C. Toher, O. Levy, K. Yang, M. Buongiorno Nardelli, and S. Curtarolo, *A RESTful API for exchanging Materials Data in the AFLOWLIB.org consortium*, Computational Materials Science **93**, 178–192 (2014). DOI: [10.1016/j.commatsci.2014.05.014](https://doi.org/10.1016/j.commatsci.2014.05.014). [229 citations.]
11. R.H. Taylor, *Materials Selection and the Stages of Design: Challenging the Convention*, Journal of Materials Education **36**(6), 139–144 (2014). [6 citations.]
- [2013]**
10. R.H. Taylor, *Crystal Symmetry Algorithms in a High-Throughput Framework for Materials*, Ph.D. Dissertation, Duke University (2013).
- [2012]**
9. S. Curtarolo, W. Setyawan, S. Wang, J. Xue, K. Yang, R.H. Taylor, L.J. Nelson, et al., *AFLOWLIB.ORG: a distributed materials properties repository from high-throughput ab initio calculations*, Computational Materials Science **58**, 227–235 (2012). [1389 citations.]
8. S. Curtarolo, W. Setyawan, G.L.W. Hart, M. Jahnátek, R.V. Chepulskii, R.H. Taylor, S. Wang, et al., *AFLOW: an automatic framework for high-throughput materials discovery*, Computational Materials Science **58**, 218–226 (2012). [1832 citations.]
7. R. Taylor, *A High-Throughput Framework for Materials Research and Space Group Determination Algorithm*, M.S. Thesis, Duke University (2012).
- [2011]**
6. R.H. Taylor, S. Curtarolo, and G.L.W. Hart, *Guiding the experimental discovery of magnesium alloys*, Physical Review B **84**(8), 084101 (2011). [71 citations.]
- [2010]**
5. R.H. Taylor, S. Curtarolo, and G.L.W. Hart, *Predictions of the Pt₈Ti phase in unexpected systems*, Journal of the American Chemical Society **132**(19), 6851–6854 (2010). [32 citations.]
4. R.H. Taylor, S. Curtarolo, and G.L.W. Hart, *Ordered Magnesium–Lithium Alloys: First-Principles Predictions*, Physical Review B **81**(2), 024112 (2010). [52 citations.]
3. G. Hart, R. Forcade, T. Kerscher, R. Taylor, L. Nelson, and A. Diaz-Ortiz, *What’s New in Cluster Expansion?*, Minerals, Metals and Materials Society / AIME (2010).

[2009 and earlier]

2. R. Taylor, S. Curtarolo, and G. Hart, *Ordered Magnesium–Lithium alloys*, APS Four Corners Section Meeting Abstracts, F7.005 (2009).
1. J. Harrison, R. Taylor, and G. Hart, *Correlations of Coupled Logistic Maps*, APS March Meeting Abstracts, Y39.011 (2008).
0. R. Taylor, *Logistic maps and synchronization*, Brigham Young University M.S. Thesis (2007). [3 citations.]

11. SELECTED CITATION METRICS

- Total citations across listed publications: ~3,800+ (Google Scholar, May 2026).
- Most-cited works: AFLOW (1,832), AFLOWLIB.ORG (1,389), RESTful API for AFLOWLIB (229), AFLOW-SYM (73), Mg alloys discovery (71), Mg–Li alloys (52), Pt₃Ti predictions (32), AFLOW fleet (40), Aluminum hydrogen generation (29).
- Co-author on the two most-cited works in the AFLOW family of computational materials publications.

12. BOARD SERVICE AND NONPROFIT LEADERSHIP

- **Gifted Music School / Peterson School of Music**, Salt Lake City, UT. 2021 – present
Vice-Chair, Board of Directors. Governance Committee Chair. Founding Chair, Endowment Committee.
 - Led the institution’s first endowment campaign.
 - Consulted on the grant-writing strategy and the grant-review process.
 - Spearheaded succession planning and the Executive Director search for the founder-led nonprofit.
- **Salt Lake City Arts Council.**
Board of Directors and Grant Committee.
 - Reviewed and scored 500+ grant applications.
 - Developed internal grant-evaluation metrics adopted by the operational committee.
- **Peterson School Conservatory Choir.** 2022 – 2026
Composer and Pianist
 - Accompanist for the conservatory choir.
 - Arranger and composer of works for solo piano, violin and piano, and string quartet.

13. ARTS: EXHIBITIONS, AWARDS, AND PUBLIC INSTALLATIONS

- UVU Woodbury Museum, Group Exhibition, Orem, Utah (2019).
- Meyer Gallery Exhibition, Park City, Utah.
- University of Utah, Gittins Gallery, Juried Works on Paper Show.
- President’s Art Show, 2nd Place Award; work acquired by Salt Lake Community College.
- Unruly Gallery, Salt Lake City — Solo Exhibition.
- Writ & Vision, Provo, Utah.
- Bountiful Davis Art Center — Solo Exhibition.
- Bountiful Davis Statewide Juried Show.
- *The Lesser Chaos*, SLCC Group Show.
- *KeyChanges* — public art installation, Abravanel Hall Plaza.
- 38th Spiritual and Religious Show, Springville Museum of Art.
- *Figuratively Speaking*, Group Show, Main Gallery, Bountiful Davis Art Center.
- Salt Lake City Public Library — Solo Exhibition, Children’s Gallery.

14. POETRY PUBLICATIONS AND RECOGNITION

- *Daughter*, Orchard's, Summer 2024 — **Pushcart Prize nominee**.
- *Turning Forty*, Tulsa Review.
- *it all seemed so real*, Suspended Magazine, Volume 2.
- *Jazz Swing*, Poor Yorick.
- *of Shakespere*, BarBar.
- Public Poetry — **Prize winner**.

15. SELECTED CONFERENCE PRESENTATIONS

- *Nanoscale Size Effect on Thermodynamic Stability of Hexavalent Chromium in Welding Fume*, FABTECH (American Welding Society), Las Vegas, NV, 2016.
- *Predictions of the Pt₈Ti phase in unexpected systems*, APS March Meeting, 2010.
- *Ordered Magnesium–Lithium alloys*, APS Four Corners Section Meeting, 2009.
- *Correlations of Coupled Logistic Maps*, APS March Meeting, 2008.

16. LANGUAGES

- English — native.
- Japanese — proficient.

Curriculum Vitæ — Richard H. Taylor, Ph.D. — updated May 13, 2026.